

MOBITIL INJECTION 20 ML VIAL

DESCRIPTION:

A clear to very pale straw-coloured solution.

COMPOSITION:

Each mL contains Dobutamine Hydrochloride BP equivalent to Dobutamine 12.5mg.

Each vial of 20 mL contains a total of Dobutamine Hydrochloride equivalent to Dobutamine 250mg.

PHARMACODYNAMICS:

Dobutamine Hydrochloride is an inotropic agent whose primary activity results from stimulation of the β receptors of the heart while producing comparatively mild chronotropic, hypertensive, arrhythmogenic, and vasodilative effects. In animal studies, dobutamine produces less increase in heart rate and less decrease in peripheral vascular resistance for a given inotropic effects than does isoprenaline.

Pretreatment of dogs with reserpine or desmethyl imipramine does not alter the actions of dobutamine. This provides strong experimental evidence that the drug is a direct agonist.

PHARMACOKINETICS:

The onset of action is within one to two minutes; however, as much as ten minutes may be required to obtain the peak effect of a particular infusion rate.

The plasma half-life of Dobutamine Hydrochloride in humans is two minutes. The major routes of metabolism are methylation of the catechol and conjugation. In human urine the major excretion products are the conjugates of dobutamine and 3-O-methyl dobutamine. The 3-O-methyl derivative of dobutamine is inactive.

In patients with depressed cardiac function, both dobutamine and isoprenaline increase the cardiac output to a similar degree. In the case of dobutamine, this increase is usually not accompanied by marked increases in heart rate (although tachycardia is occasionally observed), and the cardiac stroke volume is usually increased. In contrast, isoprenaline increases the cardiac index primarily by increasing the heart rate while stroke volume changes little or declines. Facilitation of atrioventricular conduction has been observed in human electrophysiologic studies in normal subjects and in patients with atrial fibrillation. Systemic vascular resistance is usually decreased with administration of dobutamine. Occasionally, minimal vasoconstriction has been observed.

Most clinical experience with dobutamine is short-term, up to several hours in duration. In the limited number of patients who are studied for 24, 48, and 72 hours, a persistent increase in cardiac output occurred in some, whereas the output of others returned toward base-line values. Infusions of up to 72 hours have revealed no adverse effects other than those seen with shorter infusions.

INDICATION:

Dobutamine Hydrochloride is indicated in adults who require short-term treatment of cardiac failure secondary to acute myocardial infarction, or cardiac surgery.

RECOMMENDED DOSAGE:

Administration: Because of its short half-life, Dobutamine must be administered as a continuous intravenous infusion. This is most reliably accomplished using a precision volume control intravenous infusion set. Following the initiation of a constant rate infusion or upon changing the rate, a steady-state dobutamine plasma concentration is achieved within approximately 10 min. Thus, loading doses or bolus injections are not necessary and are not recommended.

Recommended Dosage: The rate of infusion needed to increase cardiac output usually ranges from 2.5-10 $\mu\text{g/kg/min}$ (see table). Frequently, doses up to 20 $\mu\text{g/kg/min}$ are required for adequate hemodynamic improvement. On rare occasions, infusion rates up to 40 $\mu\text{g/kg/min}$ have been reported.

Drug Delivery Rate mcg/kg/min	Infusion Delivery Rate mL / kg / min		
	250 mcg/mL* (mL/kg/min)	500 mcg/mL* (mL/kg/min)	1000 mcg/mL* (mL/kg/min)
2.5	0.01	0.005	0.0025
5	0.02	0.01	0.005
7.5	0.03	0.015	0.0075
10	0.04	0.02	0.01
12.5	0.05	0.025	0.0125
15	0.06	0.03	0.015

*250mcg/mL of diluent
[250mg/1000] mg x 1000 = 250mcg/ml
1000ml/1000 ml

*500mcg/mL or 250mg/500 mL diluent
[250mg/500] mg x 1000 = 500mcg/ml
500ml/500 ml

*1000 mcg/mL or 250mg/250 mL diluent
[250mg/1000] mg x 1000 = 1000mcg/ml
250ml/250 ml

The rate of administration and the duration of therapy should be adjusted according to the patient's response, as determined by the following clinical indicators: haemodynamic parameters, such as heart rate and rhythm, arterial pressure, and whenever possible, cardiac output and measurements of ventricular filling pressures (central venous, pulmonary capillary wedge and left atrial), and signs of pulmonary congestion and organ perfusion (urine flow, skin temperature and mental status).

Concentrations up to 5000 mg/L have been administered to humans (250 mg/50 mL). The final volume administered should be determined by the fluid requirements of the patient. Rather than abruptly discontinuing therapy with dobutamine hydrochloride, it is often advisable to decrease the dosage gradually. Dosage unit: Most reports on dobutamine hydrochloride have expressed the dose in relation to body mass, eg $\mu\text{g/kg/min}$. This practice is useful to relate doses in infants and children to those in adults. Among adults, body mass has little influence on the effect of dobutamine hydrochloride; since the dosage should be titrated in each patient, adults may be easily dosed with $\mu\text{g/min}$ units.

The dosage of dobutamine hydrochloride may be initiated at 100-200 $\mu\text{g/min}$ and increased gradually to 1000-2000 $\mu\text{g/min}$ or greater, depending on the clinical and hemodynamic response of the individual patient.

Dobutamine hydrochloride injection must be diluted to at least 50 mL at the time of administration in 5% Glucose Injection or 0.9% Sodium Chloride Injection. Although chemically stable for 24 hours, prepared solutions should be used immediately after dilution.

Compatibilities: Dobutamine hydrochloride injection when diluted to 250 and 500 $\mu\text{g/mL}$ with 0.9% Sodium Chloride Injection and 5% Glucose Injection was found to be stable for 24 hours at room temperature and in the presence of fluorescent light.

Solution containing Dobutamine may exhibit a pink colour that, if present, will increase with time. This colour change is due to slight oxidation of the drug, but there is no significant loss of potency during reconstituted time periods.

ROUTE OF ADMINISTRATION:

IV infusion

CONTRAINDICATIONS:

Dobutamine Hydrochloride is contra-indicated in patients with idiopathic hypertrophic subaortic stenosis.

WARNINGS & PRECAUTIONS:

PRECAUTIONS: During the administration of Dobutamine Hydrochloride, as with any adrenergic agent, ECG and blood pressure should be continuously monitored. In addition, pulmonary artery wedge pressure and cardiac output should be monitored whenever possible to aid in the safe and effective infusion of Dobutamine Hydrochloride.

Hypovolaemia should be corrected with suitable volume expanders before treatment with Dobutamine Hydrochloride is instituted.

In patients who have atrial fibrillation with rapid ventricular response, a digitalis preparation should be used prior to instituting therapy with Dobutamine Hydrochloride.

Animals studies indicate that Dobutamine Hydrochloride may be ineffective if the patients has recently received a β -blocking drug. In such a case, the peripheral vascular resistance may increase. No improvement may be observed in the presence of marked mechanical obstruction, such as severe valvular aortic stenosis.

There is concern that any agent which increases contractile force and heart rate may increase the size of an infarction by intensifying ischaemia, but it is not known whether dobutamine does so in man. However, animal studies have shown that massive doses of 30mg/kg/min infused for 72 hours have produced irreversible myocardial damage.

Paediatric Use: The safety and efficacy of Dobutamine Hydrochloride for use in children have not been studied.

Usage in Hypotension: When hypotension is largely attributable to decreased cardiac output and coincides with an elevated ventricular filling pressure, the infusion of dobutamine HCl may help to restore the pressure. As volume is replenished in the treatment of acute hypoperfusion states and there is an increase in pulmonary wedge or central venous pressure but with no increase in cardiac output and arterial pressure, dobutamine may improve output and help restore the arterial pressure.

In general, when mean arterial blood pressure is 70mmHg in the absence of an increased ventricular filling pressure, hypovolemia may be present and would require treatment with appropriate volume repleting solution before dobutamine HCl is given.

If arterial blood pressure remains low or decreases progressively during administration of dobutamine HCl despite adequate ventricular filling pressure and cardiac output, consideration may be given to the concomitant use of a peripheral vasoconstrictor agent, eg. dopamine or norepinephrine.

Reactions at Sites of Intravenous Infusion: Phlebitis has occasionally been reported. Local inflammatory changes have been described following inadvertent infiltration. Isolated cases of cutaneous necrosis have been reported.

WARNINGS:

Increase in Heart Rate or Blood Pressure: Dobutamine Hydrochloride may cause a marked increase in heart rate or blood pressure, especially systolic pressure. Approximately 10 percent of patients in clinical studies have had rate increases of 30 beats/minute or more, and about 7.5 percent have a 50 mm Hg or greater increase in systolic pressure. Reduction of dosage usually reverses these effects promptly. Because dobutamine facilitates atrioventricular conduction, patients with atrial fibrillation are at risk of developing an exaggerated pressor response.

Ectopic Activity: Dobutamine Hydrochloride may precipitate or exacerbate ventricular ectopic activity, but it rarely has caused ventricular tachycardia.

Anaesthetics: The myocardium may be sensitized to the effect of dobutamine by cyclopropane or halogenated hydrocarbon anaesthetics, and these should be avoided.

Warning for Sodium Metabisulphite: This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

INTERACTION WITH OTHER MEDICAMENTS:

There was no evidence of drug interactions in clinical studies in which Dobutamine Hydrochloride was administered concurrently with other drugs, including digitalis preparations, frusemide, spironolactone, lignocaine, glyceryl trinitrate, isosorbide dinitrate, morphine, atropine, heparin, protamine, potassium chloride, folic acid, and paracetamol. Preliminary studies indicate that the concomitant use of dobutamine and nitroprusside results in a higher cardiac output and, usually, a lower pulmonary wedge pressure than when either drugs is used alone. β -blocker therapy, cyclopropane or halogenated hydrocarbon anaesthetics: (see Warnings and Precautions).

USE IN PREGNANCY:

Use in Pregnancy: Reproduction studies performed in rats and rabbits have revealed no evidence of impaired fertility, harm to fetus, or teratogenic effects due to dobutamine hydrochloride. Since there are no adequate and well-controlled studies in pregnant women, and since animal reproduction studies are not always predictive of human response, dobutamine hydrochloride, should not be used during pregnancy unless the potential benefits outweigh the potential risks of the fetus.

SIDE EFFECTS:

Many of the adverse effects of Dobutamine Hydrochloride are a quantitative extension of the pharmacological actions. The following adverse effects have been reported.

Increased Heart Rate, Blood Pressure, and Ventricular Ectopic Activity: A 10 to 20 mm increase in systolic blood pressure and an increase in heart rate of five to 15 beats per minute have been noted in most patients. (See Warning regarding exaggerated chronotropic and pressure effects). Approximately 5 percent of patients have had increased premature ventricular beats during infusions. These effects are dose related and their occurrence may require that the dose be reduced.

Miscellaneous Uncommon Effects: The following adverse effects have been reported in 1 to 3 percent of patients: nausea, headache, anginal pain, Non-specific chest pain, palpitations, and shortness of breath.

No abnormal laboratory values attributable to Dobutamine Hydrochloride have been observed.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdoses of dobutamine have been reported rarely. The following is provided to serve as a guide if such an overdose is encountered.

Symptoms: Toxicity from Dobutamine hydrochloride is usually due to excessive cardiac β -receptor stimulation. The duration of action of dobutamine

hydrochloride is generally short ($T_{1/2}$ =2 min) because it is rapidly metabolized by catechol-O-methyltransferase. The symptoms of toxicity may include anorexia, nausea, vomiting, tremor, anxiety, palpitations, headache, shortness of breath and anginal and nonspecific chest pain. The positive inotropic and chronotropic effects of dobutamine on the myocardium may cause hypertension, tachyarrhythmias, myocardial ischemia and ventricular fibrillation. Hypotension may result from vasodilation.

If the product is ingested, unpredictable absorption may occur from the mouth and the gastrointestinal tract.

Treatment: In managing overdosage, consider the possibility of multiple drug overdoses, interaction among drugs and unusual drug kinetics in the patient. The initial actions to be taken in a dobutamine hydrochloride overdose are discontinuing administration, establishing an airway and ensuring oxygenation and ventilation. Resuscitative measures should be initiated promptly. Severe ventricular tachyarrhythmias may be successfully treated with propranolol or lidocaine. Hypertension usually responds to a reduction in dose or discontinuation of therapy.

Protect the patient's airway and support ventilation and perfusion. If needed, meticulously monitor and maintain, within acceptable limits, the patient's vital signs, blood gases, serum electrolytes, etc. Absorption of drugs from gastrointestinal tract may be decreased by giving activated charcoal, which, in many cases, is more effective than emesis or lavage, consider charcoal instead of or in addition to gastric emptying. Repeated doses of charcoal over time may hasten elimination of some drugs that have been absorbed. Safeguard the patient's airway when employing gastric emptying or charcoal.

Forced diuresis, peritoneal dialysis, hemodialysis or charcoal hemoperfusion have not been established as beneficial for an overdose of dobutamine hydrochloride.

INCOMPATIBILITIES:

Do not add Dobutamine Hydrochloride Injection to 5% sodium bicarbonate injection or any other strongly alkaline solutions. Dobutamine Hydrochloride should not be used in conjunction with other agents or diluent containing sodium bisulphites and ethanol.

STORAGE CONDITIONS:

Store below 30°C in cool dry place. Protect from light. Do not freeze. Keep out of reach of children. *Jauhkan dari kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Pack in 10's vials of 20 mL/ box

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59, Kawasan 3, Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.